NetRCA

Telecom Network Outage Root-Cause Analysis Assistant

Project Status Report

PHASE 1	PHASE 2	NEXT
COMPLETE	COMPLETE	PHASE 3

Prepared for TPM Review

UST Enterprise Capstone - Telecom Domain

Current status: deterministic incident-processing foundation and RAG retrieval layer completed; AI reasoning and orchestration are the next development stages.

1. Executive Summary

NetRCA is a telecom outage investigation assistant designed to help NOC engineers move from raw outage symptoms to an evidence-backed root-cause analysis (RCA), recommended remediation, and a human-approved incident report.

The project follows a controlled enterprise workflow: incident input and logs are validated and normalized; multiple evidence sources are analyzed; relevant troubleshooting knowledge is retrieved through RAG; AI reasoning agents will synthesize the probable root cause and remediation; deterministic confidence and routing logic will decide the level of review; and the final decision remains with a human engineer.

Overall end goal
Help a NOC engineer understand what happened, why it happened, what evidence supports the conclusion, what should be done next, and approve the final action/report.

2. Business Problem

- During a network outage, evidence is distributed across alarms, event logs, KPI degradation, historical incidents, and troubleshooting SOPs.
- Manual correlation of these sources can be time-consuming and depends heavily on engineer experience.
- A useful enterprise solution must therefore be explainable: the system should show the evidence behind its conclusion instead of returning an unsupported chatbot answer.

3. Proposed Solution and Target Workflow

Design principle: use deterministic logic where consistency and auditability matter, and AI where reasoning across evidence adds value.

Step	Stage	Purpose
1	Incident + Logs	Outage ticket, alarms, KPI events and engineer notes
2	Validate / Parse	Check input quality and normalize technical logs
3	Parallel Analysis	Alarm, KPI and historical-pattern analysis
4	RAG Retrieval	Retrieve relevant telecom SOP/reference chunks
5	RCA + Confidence	Synthesize probable root cause and calculate explainable confidence
6	Router + Recommendation	Determine review type and propose grounded remediation
7	Human Approval	Engineer approves, rejects or requests revision
8	Final Report	Generate traceable incident summary and recovery plan

4. Phase-wise Progress

Phase 1 - Synthetic Data and Deterministic Tools

Status: COMPLETE

Completed Item	Outcome
Synthetic incidents	12 telecom incident records covering valid and error scenarios.
Network logs	231 alarm/event/KPI log rows with useful evidence and distractor/noisy events.
Historical incidents	24 resolved incident records with known root causes and resolutions.
Input Validation	Checks required fields, data types, timestamps and allowed values before analysis.
Log Parser	Normalizes timestamps/log levels, separates alarms from KPI rows and extracts error codes.
Severity Calculator	Deterministic P1-P4 severity using customer impact, service impact and alarm priority, with an explainable breakdown.
Incident Lookup	Returns top historical matches using explainable keyword, device-type and alarm-family similarity.
Testing	Phase 1 was reported with 227 unit tests passing in the project environment.

Why Phase 1 matters: it establishes a reliable, explainable data foundation before introducing LLM reasoning.

Phase 2 - Knowledge Base and Document Retrieval (RAG Foundation)

Status: COMPLETE

Completed Item	Outcome
Knowledge base	12 synthetic telecom SOP/reference documents covering link failures, packet loss, congestion, DNS, BGP, power, QoS, alarm and KPI guidance.
Document Loader	Loads and validates document front matter and source metadata.
Chunking	Header-aware chunking with configurable size/overlap to preserve useful troubleshooting context.
Embeddings	Sentence Transformers all-MiniLM-L6-v2, producing 384-dimensional semantic vectors.
Vector Database	Persistent local ChromaDB collection for semantic similarity search.

Completed Item	Outcome
Retriever	Returns top relevant chunks with source document, section, chunk ID and retrieval score; supports weak-retrieval detection.
Retrieval Index	12 documents produced 105 retrieval chunks in the completed Phase 2 build.
Testing	Combined project status was reported as 327 tests passing after Phase 2 completion.

Why Phase 2 matters: the later RCA and Recommendation agents can retrieve trusted troubleshooting context instead of reasoning only from the model's internal knowledge.

5. Current Demonstrable Capability

- Load a synthetic telecom incident and validate whether its structure is acceptable.
- Parse and normalize its raw alarms/events/KPI records.
- Calculate an explainable P1-P4 severity.
- Find similar previously resolved incidents.
- Search the telecom knowledge base semantically and return traceable SOP/reference chunks.
- Detect weak retrieval when the knowledge base does not contain sufficiently relevant evidence.

6. Remaining Development

Phase	Scope	Planned Deliverable	Status
Phase 3	LLM Reasoning Agents	Build and independently test Alarm Analysis, KPI Analysis, Historical Pattern, RCA and Recommendation agents using structured outputs.	NEXT
Phase 4	LangGraph Orchestration	Connect the workflow using shared state, sequential + parallel execution, retrieval, deterministic routing and human-interrupt flow.	PENDING
Phase 5	Reliability & Observability	Add deterministic confidence calculation, retries, fallbacks, structured tracing and JSONL logging.	PENDING
Phase 6	Streamlit UI	Create the demo interface for incident selection, evidence, RCA, recommendations, review and report generation.	PENDING
Phase 7	End-to-End Testing	Run representative success, low-confidence, missing-data, weak-retrieval, high-severity and revision scenarios.	PENDING
Phase 8	Demo / Documentation	Finalize demo script, architecture explanation, Q&A preparation and project documentation.	PENDING

7. Final System Behaviour

Target evaluator/demo experience: select an outage -> view normalized evidence -> observe parallel analysis -> view retrieved SOP evidence -> receive evidence-backed RCA and confidence -> review remediation -> approve/reject/revise -> generate final incident report.

8. Key Design Decisions

- **Deterministic before AI:** Validation, parsing, severity, historical lookup, routing and confidence are rule-based for consistency and explainability.
- **AI only for reasoning:** LLM agents are reserved for interpreting alarms/KPIs, synthesizing RCA and generating evidence-grounded remediation.
- **RAG for grounding:** Recommendations and RCA reasoning will be supported by retrieved telecom SOP/reference chunks.
- **Mandatory human approval:** Every successful RCA path reaches a human review checkpoint; the router controls review type, not whether approval is required.
- **Local-first implementation:** The capstone is designed to run locally; cloud deployment is not required for the core demo.

9. Dependencies / Open Items

- LLM provider/API credentials for Phase 3 must be confirmed in the approved company environment before agent integration.
- Retrieval thresholds and chunking parameters remain configurable and should be tuned only if standalone retrieval tests show a need.
- Phase 2 completion and test totals reflect the project's reported execution environment; this report does not claim independent re-execution of all dependencies outside that environment.

10. Current Status Summary

Completed	Architecture freeze, Phase 1 deterministic foundation, Phase 2 RAG/retrieval foundation
Current capability	Validated incident processing + historical similarity + semantic retrieval from telecom knowledge base
Next immediate milestone	Phase 3 - build and independently test the five LLM reasoning agents
Final outcome	Evidence-backed RCA and remediation with deterministic routing, mandatory human approval, and a downloadable final incident report

Overall status: Phase 1 and Phase 2 completed; reasoning, orchestration, reliability, UI and final demo integration remain.